AI-Powered Automated Web Penetration Testing & Security Hardening Platform

Course Code: CSE-3200

Course Title: Software Development Project (Phase-I)

INTRODUCTION

Developing an AI-assisted web security assessment platform that analyzes authorized web applications, identifies security vulnerabilities, explains risks, and provides security hardening recommendations for enhanced protection.

OBJECTIVES

- To scan real websites and identify security vulnerabilities
- To detect common web attacks (SQLi, XSS, CSRF, IDOR, etc.)
- To analyze the risk level and business impact of each vulnerability
- To recommend effective security hardening solutions
- To generate professional security assessment reports

EXPECTED OUTCOMES

- Risk-Based Prioritization
- Security Posture Tracking
- Automated Vulnerability Discovery
- Verification & Rescan
- Cost & Time Savings

REFERENCE

M. Y. Darus, M. F. B. Bolhan, A. Kurniawan, Y. Muliono, C. R. Pardomuan, and M. M. Hata, "Enhancing web application penetration testing with a static application security testing (SAST) tool," in Proc. 2023 IEEE 8th International Conference on Recent Advances and Innovations in Engineering (ICRAIE), 2023, pp. 1–6.

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WORKFLOW DIAGRAM

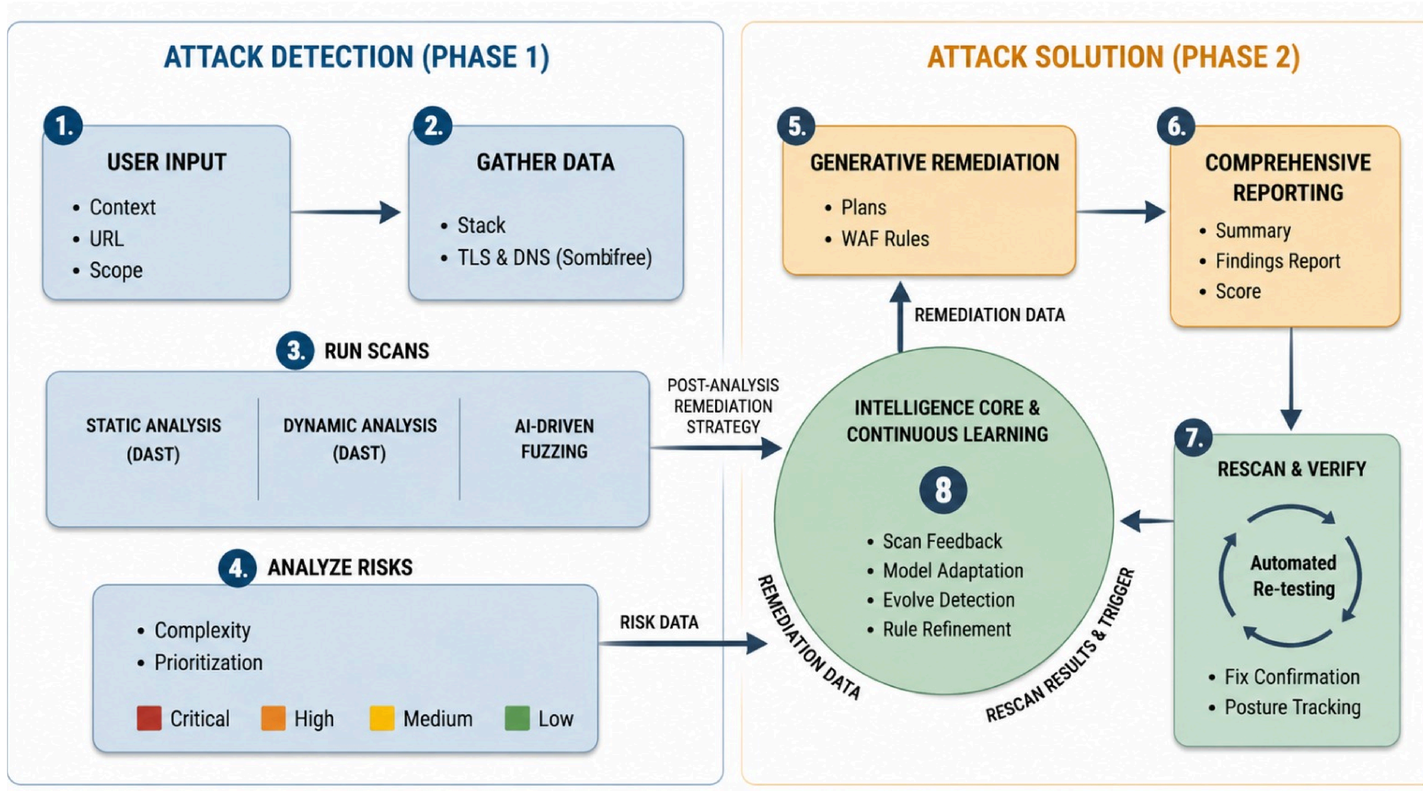


Figure- 1: End-to-End Automated Security Testing & Remediation Workflow

KEY FEATURES

🌐 REAL WEBSITE SECURITY SCANNING
🛡️ ATTACK DETECTION
📊 RISK LEVEL CLASSIFICATION
🛠️ HARDENING SOLUTIONS
📋 SCAN HISTORY & COMPARISON

FUTURE WORKS

- Authentication Bypass Testing
- API Security Scanning
- Real-Time Browser Extension
- Continuous Monitoring

CONCLUSION

The AI-Powered Automated Web Penetration Testing & Security Hardening Platform is a revolutionary solution designed to transform how web application security is assessed and improved. This project successfully combines automated vulnerability scanning, AI-driven analysis, and actionable hardening recommendations into a single, easy-to-use platform.